

Minimal acyclic forbidden minors for the family of graphs with bounded path-width

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Abstract

The graphs with bounded path-width, introduced by Robertson and Seymour, and the graphs with bounded proper-path-width, introduced in this paper, are investigated. These families of graphs are minor-closed. We characterize the minimal acyclic forbidden minors for these families of graphs. We also give estimates for the numbers of minimal forbidden minors and for the numbers of vertices of the largest minimal forbidden minors for these families of graphs.

1. Introduction

Graphs we consider are finite and undirected, but may have loops and multiple edges unless otherwise specified. A graph H is a minor of a graph G if H is isomorphic to a graph obtained from a subgraph of G by contracting edges. A family $\mathcal{F}$ of graphs is said to be minor-closed if the following condition holds: If $G \in \mathcal{F}$ and H is a minor of G then $H \in \mathcal{F}$. A graph G is a minimal forbidden minor for a minor-closed family $\mathcal{F}$ of graphs if $G \notin \mathcal{F}$ and any proper minor of G is in $\mathcal{F}$.

Robertson and Seymour proved the following deep theorems.

Theorem 1.1 (Robertson and Seymour [15]). *Every minor-closed family of graphs has a finite number of minimal forbidden minors.*

Theorem 1.2 (Robertson and Seymour [14]). *The problem of deciding if a fixed graph is a minor of an input graph can be solved in polynomial time.*

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The combination of these theorems suggests the existence of a polynomial-time algorithm for the problem of testing membership for any minor-closed family $\mathcal{F}$ of graphs. Although many important problems are known to be reduced to the problem, we cannot have a polynomial-time algorithm unless we can find all the minimal forbidden minors for $\mathcal{F}$. Unfortunately, it was proved that there is no general method to find all the minimal forbidden minors for any minor-closed family of graphs [6, 7], as suspected. However, special arguments could be applied for individual minor-closed family. In fact, the minimal forbidden minors were found for families of planar graphs [21], graphs embeddable on the projective plane [9, 1], partial 2-trees [3], partial 3-trees [3, 4], and graphs with path-width at most 2 [5].

We investigate the family $\mathcal{F}_k$ of graphs with path-width at most k for any $k \geq 0$. We introduce the proper-path-width of graphs and investigate the family $\mathcal{P}_k$ of graphs with proper-path-width at most k for any $k \geq 1$. $\mathcal{F}_k$ and $\mathcal{P}_k$ are minor-closed families. We show that every minimal acyclic forbidden minor for $\mathcal{F}_k$ (resp. $\mathcal{P}_k$) can be obtained from those for $\mathcal{F}_{k-1}$ (resp. $\mathcal{P}_{k-1}$) by a simple composition. We also give estimates for the numbers of minimal forbidden minors for $\mathcal{F}_k$ and $\mathcal{P}_k$, and the numbers of vertices of the largest minimal forbidden minors for $\mathcal{F}_k$ and $\mathcal{P}_k$.

2. Minimal acyclic forbidden minors for graphs with bounded path-width

The path-width of a graph was introduced by Robertson and Seymour [13]. We denote the vertex set and edge set of a graph G by $V(G)$ and $E(G)$, respectively.

Definition 2.1. Let $\mathcal{X} = (X_1, X_2, \dots, X_r)$ be a sequence of subsets of $V(G)$. The *width* of $\mathcal{X}$ is $\max_{1 \leq i \leq r} |X_i| - 1$. $\mathcal{X}$ is called a *path-decomposition* of G if the following conditions are satisfied:

- (i) $\bigcup_{1 \leq i \leq r} X_i = V(G)$;
- (ii) for any edge $(u, v) \in E(G)$, there exists an i such that $u, v \in X_i$;
- (iii) for all l, m , and n with $1 \leq l \leq m \leq n \leq r$, $X_l \cap X_n \subseteq X_m$.

The *path-width* of G , denoted by $\text{pw}(G)$, is the minimum width over all path-decompositions of G .

For a sequence $\mathcal{X} = (X_1, X_2, \dots, X_r)$ of subsets of $V(G)$ and a set $S \subseteq V(G)$, we denote the sequence $(X_1 \cup S, X_2 \cup S, \dots, X_r \cup S)$ by $(\mathcal{X} \cup S)$, and $(X_1 \cap S, X_2 \cap S, \dots, X_r \cap S)$ by $(\mathcal{X} \cap S)$ for simplicity.

The following lemmas mentioned in [13] will be used later.

Lemma 2.2 (Robertson and Seymour [13]). *If every connected component of G has path-width $\leq k$, then G has path-width $\leq k$.*

Lemma 2.3 (Robertson and Seymour [13]). *If $S \subseteq V(G)$ and $\text{pw}(G \setminus S) \leq k$, then $\text{pw}(G) \leq k + |S|$, where $G \setminus S$ denotes the graph obtained from G by deleting the vertices in S .*

Let $\mathcal{F}_k$ be the family of graphs with path-width at most k . It is easy to see that $\mathcal{F}_k$ is minor-closed. Let $\Omega(\mathcal{F}_k)$ be the set of all minimal forbidden minors for $\mathcal{F}_k$, and $\Omega_a(\mathcal{F}_k)$ be the set of all minimal acyclic forbidden minors for $\mathcal{F}_k$. Obviously, $\Omega_a(\mathcal{F}_k) \subseteq \Omega(\mathcal{F}_k)$.

We introduce the star-composition of graphs which plays an important role in the following.

Definition 2.4. Let H_1, H_2 , and H_3 be graphs. A graph obtained from H_1, H_2 , and H_3 by the following construction is called a *star-composition* of H_1, H_2 , and H_3 :

- (i) choose a vertex $v_i \in V(H_i)$ for $i=1, 2$, and 3 ;
- (ii) let v be a new vertex not in H_1, H_2 , or H_3 ;
- (iii) connect v to v_i by an edge (v, v_i) for $i=1, 2$, and 3 .

The vertex v is called the *center* of the star-composition.

We first prove the following theorem.

Theorem 2.5. Let $k \geq 1$. A tree T is in $\Omega_a(\mathcal{F}_k)$ if and only if T is a star-composition of (not necessarily distinct) three trees in $\Omega_a(\mathcal{F}_{k-1})$.

We prove this theorem by a series of lemmas.

Lemma 2.6. Let $(X_1, X_2, \dots, X_r)$ be a path-decomposition of a graph G and $v \in V(G)$. If $v \in X_i$ and $v \in X_j$ ($1 \leq i \leq j \leq r$), then $v \in X_l$ for any l ($i \leq l \leq j$). That is, v appears in consecutive X_i 's.

Proof. Suppose that $v \notin X_l$ for some l ($i < l < j$). Then $X_i \cap X_j \not\subseteq X_l$, contradicting Definition 2.1(iii). $\square$

Lemma 2.7. Let H be a connected subgraph of a graph G , and $(X_1, X_2, \dots, X_r)$ be a path-decomposition of G . If $X_i \cap V(H) \neq \emptyset$ and $X_j \cap V(H) \neq \emptyset$ ($1 \leq i \leq j \leq r$), then $X_l \cap V(H) \neq \emptyset$ for any l ($i \leq l \leq j$), i.e. the vertices of H appear in consecutive X_i 's.

Proof. Suppose that $X_l \cap V(H) = \emptyset$ for some l ($i < l < j$). Each vertex of H appears in consecutive X_i 's by Lemma 2.6. Thus, if $P = \bigcup_{i=1}^{l-1} (X_i \cap V(H))$ and $Q = \bigcup_{i=l+1}^r (X_i \cap V(H))$, $P \cap Q = \emptyset$. Since $V(H)$ is partitioned into P and Q , and H is connected, there exist $u \in P$ and $v \in Q$ such that $(u, v) \in E(H)$. However, $\{u, v\} \not\subseteq X_i$ for any i ($1 \leq i \leq r$), contradicting Definition 2.1(ii). $\square$

Lemma 2.8. Let G be a connected graph and k be a positive integer. If G has a vertex v such that $G \setminus \{v\}$ has at least three connected components with path-width k or more, then $\text{pw}(G) \geq k+1$.

Proof. We may assume that the path-widths of connected components of $G \setminus \{v\}$ are at most k , for otherwise trivially $\text{pw}(G) \geq k+1$. Let H_1, H_2 , and H_3 be connected

components of $G \setminus \{v\}$ with path-width k , and $v_1 \in V(H_1)$, $v_2 \in V(H_2)$, and $v_3 \in V(H_3)$ be vertices adjacent to v in G .

Suppose contrary that $\text{pw}(G) \leq k$ and G has a path-decomposition $\mathcal{X} = (X_1, X_2, \dots, X_r)$ with width $\leq k$. There exists some i_j such that $X_{i_j} \subseteq V(H_j)$ for $j=1, 2$, and 3 , for otherwise $(\mathcal{X} \cap V(H_j))$ is a path-decomposition of H_j with width $\leq k-1$ and $\text{pw}(H_j) \leq k-1$. Without loss of generality, we assume that $i_1 < i_2 < i_3$. It is trivial that $G \setminus V(H_2)$ is a connected subgraph of G . However, $X_{i_1} \cap V(G \setminus V(H_2)) \neq \emptyset$ and $X_{i_2} \cap V(G \setminus V(H_2)) = \emptyset$, and $X_{i_3} \cap V(G \setminus V(H_2)) \neq \emptyset$, contradicting Lemma 2.7. Thus, $\text{pw}(G) \geq k+1$. $\square$

Lemma 2.9. *Let T be a tree and k be a positive integer. Suppose that for any $v \in V(T)$, $T \setminus \{v\}$ has no connected component with path-width $k+1$ or more and at most two connected components with path-width k . Then $\text{pw}(T) \leq k$.*

Proof. Let T_0 be T , and let v_0 be a vertex such that $T_0 \setminus \{v_0\}$ has the maximum number of connected components with path-width k .

If $T_0 \setminus \{v_0\}$ has no connected component with path-width k , then $\text{pw}(T) \leq k$ by Lemmas 2.2 and 2.3.

If $T_0 \setminus \{v_0\}$ has two connected components with path-width k , let T_1 be one of these components and $v_1 \in V(T_1)$ be a vertex adjacent to v_0 in T_0 . We recursively define T_i and $v_i \in V(T_i)$ ($1 \leq i \leq a$) while $T_{i-1} \setminus \{v_{i-1}\}$ has a component with path-width k as follows: Let T_i be a connected component of $T_{i-1} \setminus \{v_{i-1}\}$ with path-width k and $v_i \in V(T_i)$ be a vertex adjacent to v_{i-1} in T_{i-1} . $T_a \setminus \{v_a\}$ has no connected component with path-width k . Let T_{a+1} be the other connected component of $T_0 \setminus \{v_0\}$ with path-width k , and $v_{a+1} \in V(T_{a+1})$ be a vertex adjacent to v_0 in T_0 . Define recursively T_i and $v_i \in V(T_i)$ ($a+1 \leq i \leq b$) as above. Notice that $T_i \setminus \{v_i\}$ ($1 \leq i \leq b$) has at most one connected component with path-width k , for otherwise $T_0 \setminus \{v_i\}$ has three or more connected components with path-width k , contradicting the assumption of the lemma.

Let H'_i ($0 \leq i \leq b$) be the union of components of $T_i \setminus \{v_i\}$ with path-width $\leq k-1$, and H_i ($0 \leq i \leq b$) be the induced subgraph of T on $V(H'_i) \cup \{v_i\}$. By Lemma 2.2, $\text{pw}(H'_i) \leq k-1$ ($0 \leq i \leq b$). By Lemma 2.3, $\text{pw}(H_i) \leq k$ ($0 \leq i \leq b$). Let $\mathcal{X}'_{(i)}$ be a path-decomposition of H'_i with width $\leq k-1$. Then $\mathcal{X}_{(i)} = (\mathcal{X}'_{(i)} \cup \{v_i\})$ is a path-decomposition of H_i with width $\leq k$. We define sequences $\mathcal{L}$ and $\mathcal{R}$ as follows:

$$\begin{aligned} \mathcal{L} &= (\mathcal{X}_{(a)}, \{v_a, v_{a-1}\}, \mathcal{X}_{(a-1)}, \{v_{a-1}, v_{a-2}\}, \dots, \mathcal{X}_{(2)}, \{v_2, v_1\}, \mathcal{X}_{(1)}, \{v_1, v_0\}), \\ \mathcal{R} &= (\{v_0, v_{a+1}\}, \mathcal{X}_{(a+1)}, \{v_{a+1}, v_{a+2}\}, \\ &\quad \mathcal{X}_{(a+2)}, \dots, \{v_{b-2}, v_{b-1}\}, \mathcal{X}_{(b-1)}, \{v_{b-1}, v_b\}, \mathcal{X}_{(b)}). \end{aligned}$$

It is easy to see that the sequence $(\mathcal{L}, \mathcal{X}_{(0)}, \mathcal{R})$ is a path-decomposition of T and $\text{pw}(T) \leq k$.

If $T_0 \setminus \{v_0\}$ has just one connected component with path-width k , the sequence $\mathcal{R}$ above is empty, and $(\mathcal{L}, \mathcal{X}_{(0)})$ is a path-decomposition of T and we also have $\text{pw}(T) \leq k$. $\square$

Lemma 2.10. *For any tree T and integer $k \geq 1$, $\text{pw}(T) \geq k+1$ if and only if T has a vertex v such that $T \setminus \{v\}$ has at least three connected components with path-width k or more.*

Proof. Suppose that $\text{pw}(T) \geq k+1$. Let T' be a minimal subgraph of T with $\text{pw}(T') \geq k+1$. Since T' is minimal, $T' \setminus \{w\}$ has no connected component with path-width $\geq k+1$ for any $w \in V(T')$. Thus, there exists a vertex $v \in V(T')$ such that $T' \setminus \{v\}$ has at least three connected components with path-width k , for otherwise $\text{pw}(T') \leq k$ by Lemma 2.9. Hence, $T \setminus \{v\}$ has at least three connected components with path-width $\geq k$. The converse follows from Lemma 2.8. $\square$

Lemma 2.11. *If $k \geq 1$ and H_1, H_2 , and H_3 are (not necessarily distinct) graphs in $\Omega(\mathcal{F}_{k-1})$ then any star-composition of H_1, H_2 , and H_3 is in $\Omega(\mathcal{F}_k)$.*

Proof. Let G be a star-composition of H_1, H_2 , and H_3 , and v be the center of the star-composition. Since $H_i \in \Omega(\mathcal{F}_{k-1})$ ($i=1, 2$, and 3), $G \setminus \{v\}$ has three connected components with path-width k . Thus, $\text{pw}(G) \geq k+1$ by Lemma 2.8. On the other hand, $\text{pw}(G \setminus \{v\}) \leq k$ by Lemma 2.2, and so $\text{pw}(G) \leq k+1$ by Lemma 2.3. Hence, we have $\text{pw}(G) = k+1$.

Next we show that G is minimal. Let $v_i \in V(H_i)$ be a vertex adjacent to v in G . Since $H_i \in \Omega(\mathcal{F}_{k-1})$, $\text{pw}(H_i \setminus \{v_i\}) = k-1$. Let $\mathcal{X}_{(i)}$ be a path-decomposition of $H_i \setminus \{v_i\}$ with width $k-1$ ($i=1, 2$, and 3). It is sufficient to show that the path-width of a minor G' of G obtained by deleting or contracting an edge e is at most k .

Case 1: $e \in \{(v, v_1), (v, v_2), (v, v_3)\}$.

Without loss of generality, we assume that $e = (v, v_1)$. If G' is obtained by deleting edge (v, v_1) , the sequence $(\mathcal{X}_{(1)} \cup \{v_1\}, \mathcal{X}_{(2)} \cup \{v_2\}, \{v, v_2\}, \{v, v_3\}, \mathcal{X}_{(3)} \cup \{v_3\})$ is a path-decomposition of G' and $\text{pw}(G') \leq k$. If G' is obtained by contracting edge (v, v_1) , the sequence $(\mathcal{X}_{(2)} \cup \{v_2\}, \{v, v_2\}, \mathcal{X}_{(1)} \cup \{v\}, \{v, v_3\}, \mathcal{X}_{(3)} \cup \{v_3\})$ is a path-decomposition of G' and again $\text{pw}(G') \leq k$.

Case 2: $e \notin \{(v, v_1), (v, v_2), (v, v_3)\}$.

Without loss of generality, we assume that $e \in E(H_2)$. G' is a star-composition of H_1, H'_2 , and H_3 , where H'_2 is a minor of H_2 obtained by deleting or contracting e . Let $\mathcal{X}'$ be a path-decomposition of H'_2 with width $\leq k-1$. Then the sequence $(\mathcal{X}_{(1)} \cup \{v_1\}, \{v, v_1\}, \mathcal{X}' \cup \{v\}, \{v, v_3\}, \mathcal{X}_{(3)} \cup \{v_3\})$ is a path-decomposition of G' and $\text{pw}(G') \leq k$.

Thus, the path-width of any proper minor of G is at most k , and G is minimal.

Hence $G \in \Omega(\mathcal{F}_k)$. $\square$

Corollary 2.12. *If $k \geq 1$ and T_1, T_2 , and T_3 are (not necessarily distinct) trees in $\Omega_a(\mathcal{F}_{k-1})$ then any star-composition of T_1, T_2 , and T_3 is in $\Omega_a(\mathcal{F}_k)$.*

Proof. Any star-composition of trees is also a tree. $\square$

Lemma 2.13. *If $k \geq 1$ and T is any tree $\Omega_a(\mathcal{F}_k)$ then T is a star-composition of some (not necessarily distinct) trees T_1, T_2 , and T_3 in $\Omega_a(\mathcal{F}_{k-1})$.*

Proof. There exists a vertex v such that $T \setminus \{v\}$ has three or more connected components with path-width k or more by Lemma 2.10. Because T is minimal, $T \setminus \{v\}$ has exactly three connected components with path-width k . Let T_1, T_2 , and T_3 be connected components of $T \setminus \{v\}$. Suppose $T_1 \notin \Omega_a(\mathcal{F}_{k-1})$. Let T'_1 be a proper minor of T_1 with path-width k and T' be a star-composition of T'_1, T_2 , and T_3 . Then $\text{pw}(T') = k + 1$, contradicting that $T \in \Omega_a(\mathcal{F}_k)$. Thus, $T_1 \in \Omega_a(\mathcal{F}_{k-1})$. Similarly, T_2 and T_3 are in $\Omega_a(\mathcal{F}_{k-1})$. $\square$

By Corollary 2.12 and Lemma 2.13, we obtain Theorem 2.5.



Fig. 1. The graphs in $\Omega(\mathcal{F}_1)$.

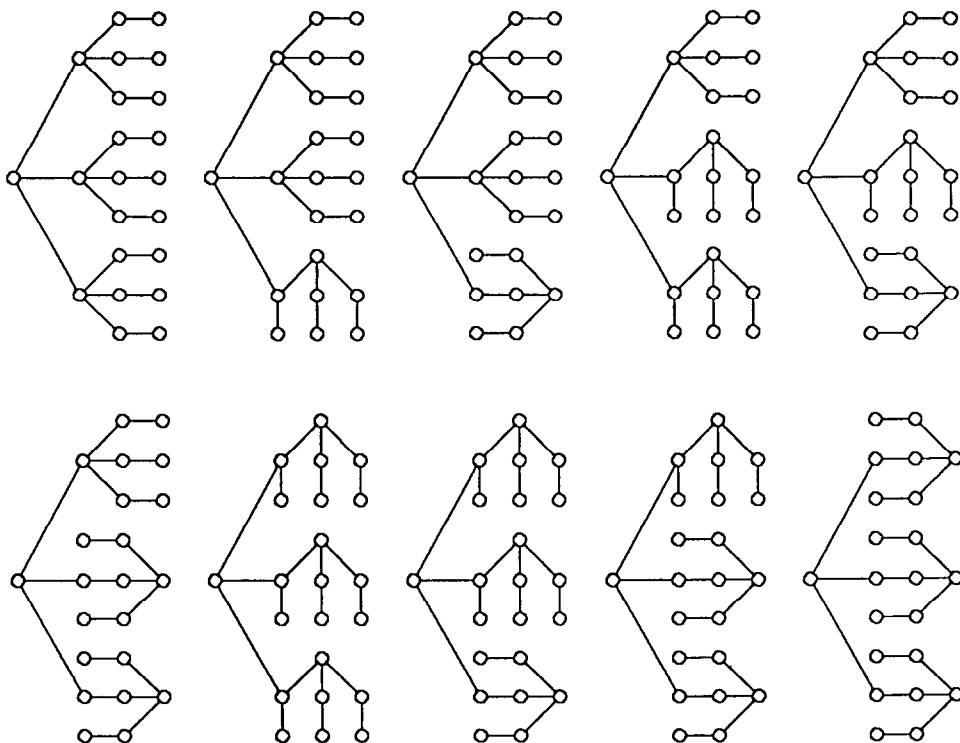


Fig. 2. The trees in $\Omega_a(\mathcal{F}_2)$.

It is easy to see that $\Omega(\mathcal{F}_0) = \{K_2\}$. The graphs in $\Omega(\mathcal{F}_1)$ and $\Omega_a(\mathcal{F}_2)$ are shown in Figs. 1 and 2, respectively. The following corollary can be easily proved by induction on k .

Corollary 2.14. (1) *The number of vertices of a tree in $\Omega_a(\mathcal{F}_k)$ is $(5 \cdot 3^k - 1)/2$ ($k \geq 0$).*
 (2) $|\Omega_a(\mathcal{F}_k)| \geq k!^2$ ($k \geq 0$).

We counted $|\Omega_a(\mathcal{F}_k)|$ for $k=0, 1, 2, 3$, and 4 as follows: $|\Omega_a(\mathcal{F}_0)| = |\Omega_a(\mathcal{F}_1)| = 1$, $|\Omega_a(\mathcal{F}_2)| = 10$, $|\Omega_a(\mathcal{F}_3)| = 117, 480$, $|\Omega_a(\mathcal{F}_4)| = 14, 403, 197, 619, 396, 707, 660$.

3. Minimal acyclic forbidden minors for graphs with bounded proper-path-width

We introduce in this section the proper-path-width of graphs.

Definition 3.1. The path-decomposition $(X_1, X_2, \dots, X_r)$ of G with width k ($k \geq 1$) is called a *proper-path-decomposition* of G if the following condition holds: For any X_l, X_m , and X_n such that each one is not a subset of the others ($1 \leq l < m < n \leq r$), $|X_l \cap X_m \cap X_n| \leq k - 1$. The *proper-path-width* of G , denoted by $ppw(G)$, is the minimum width over all proper-path-decompositions of G .

Let $\mathcal{P}_k$ be the family of graphs with proper-path-width at most k . It is easy to see that $\mathcal{P}_k$ is minor-closed. Let $\Omega(\mathcal{P}_k)$ be the set of all minimal forbidden minors for $\mathcal{P}_k$, and $\Omega_a(\mathcal{P}_k)$ be the set of all minimal acyclic forbidden minors for $\mathcal{P}_k$.

A k -clique of a graph G is a complete subgraph of G on k vertices. For a positive integer k , k -trees are defined recursively as follows: (1) The complete graph on k vertices is a k -tree; (2) Given a k -tree Q on n vertices ($n \geq k$), a graph obtained from Q by adding a new vertex adjacent to the vertices of a k -clique of Q is a k -tree on $n + 1$ vertices. A k -tree Q is called a k -path [12] or k -chordal path [2] if $|V(Q)| \leq k + 1$ or Q has exactly two vertices of degree k . A k -separator S of a k -tree Q is a k -clique of Q such that $Q \setminus V(S)$ has at least two connected components. For a positive integer k , interior k -caterpillars [12] are defined as follows: (1) A k -path is an interior k -caterpillar; (2) Given an interior k -caterpillar Q on n vertices ($n \geq k + 2$), a graph obtained from Q by adding a new vertex adjacent to the vertices of a k -separator of Q is also an interior k -caterpillar on $n + 1$ vertices.

A 1-path, interior 1-caterpillar, and 1-tree are an ordinary path, caterpillar, and tree, respectively. A subgraph of a k -path and interior k -caterpillar are called a *partial k -path* and *partial interior k -caterpillar*, respectively.

It is not difficult to see that, for any simple graph G , $pw(G) \leq k$ ($k \geq 1$) if and only if G is a partial interior k -caterpillar [19], and $ppw(G) \leq k$ ($k \geq 1$) if and only if G is a partial k -path [17].

We have the following lemma and theorem for $\mathcal{P}_k$ corresponding to Lemma 2.11 and Theorem 2.5, respectively. Proofs are almost same as those for $\mathcal{F}_k$, and are omitted.

Lemma 3.2. *If $k \geq 2$ and H_1, H_2 , and H_3 are (not necessarily distinct) graphs in $\Omega(\mathcal{P}_{k-1})$ then any star-composition of H_1, H_2 , and H_3 is in $\Omega(\mathcal{P}_k)$.*

Theorem 3.3. *Let $k \geq 2$. A tree T is in $\Omega_a(\mathcal{P}_k)$ if and only if T is a star-composition of (not necessarily distinct) three trees in $\Omega_a(\mathcal{P}_{k-1})$.*

It is easy to see that $\Omega(\mathcal{P}_1) = \{K_3, K_{1,3}\}$. The trees in $\Omega_a(\mathcal{P}_2)$ are shown in Fig. 3.

Corollary 3.4. (1) *The number of vertices of a tree in $\Omega_a(\mathcal{P}_k)$ is $(3^{k+1} - 1)/2$ ($k \geq 1$).*
 (2) $|\Omega_a(\mathcal{P}_k)| \geq k!^2$ ($k \geq 1$).

We counted $|\Omega_a(\mathcal{P}_k)|$ for $k=1, 2, 3$, and 4 as follows: $|\Omega_a(\mathcal{P}_1)|=1$, $|\Omega_a(\mathcal{P}_2)|=4$, $|\Omega_a(\mathcal{P}_3)|=1,330$, $|\Omega_a(\mathcal{P}_4)|=2,875,919,312,080$.

Another kind of composition is possible for $\Omega(\mathcal{P}_k)$.

Definition 3.5. A *delta-composition* of graphs H_1, H_2 , and H_3 is a graph obtained from H_1, H_2 , and H_3 by the following construction:

- (i) choose a vertex $v_i \in V(H_i)$ for $i=1, 2$, and 3;
- (ii) connect v_1 to v_2, v_2 to v_3 , and v_3 to v_1 by edges $(v_1, v_2), (v_2, v_3)$, and (v_3, v_1) , respectively.

Theorem 3.6. *If $k \geq 2$ and H_1, H_2 , and H_3 are (not necessarily distinct) graphs in $\Omega(\mathcal{P}_{k-1})$ then any delta-composition of H_1, H_2 , and H_3 is in $\Omega(\mathcal{P}_k)$.*

Proof. Let G be a delta-composition of H_1, H_2 , and H_3 . Let $v_i \in V(H_i)$ be the chosen vertex for $i=1, 2$, and 3. Because $H_i \in \Omega(\mathcal{P}_{k-1})$, $\text{ppw}(H_i \setminus \{v_i\}) = k-1$. Let $\mathcal{X}_{(i)}$ be the proper-path-decomposition of $H_i \setminus \{v_i\}$ with width $k-1$ ($i=1, 2$, and 3).

First, we show that $\text{ppw}(G) = k+1$. It is easy to see that the sequence $(\mathcal{X}_{(1)} \cup \{v_1\}, \mathcal{X}_{(2)} \cup \{v_1, v_2\}, \{v_1, v_2, v_3\}, \mathcal{X}_{(3)} \cup \{v_3\})$ is a proper-path-decomposition of

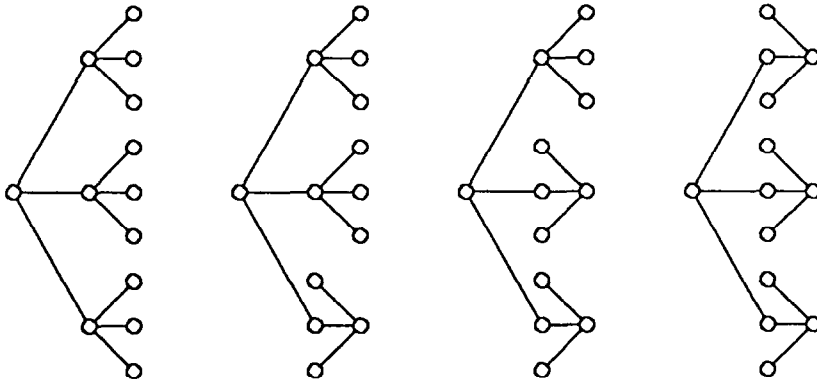


Fig. 3. The trees in $\Omega_a(\mathcal{P}_2)$.

G , and $\text{ppw}(G) \leq k+1$. Suppose that $\text{ppw}(G) \leq k$ and G has a proper-path-decomposition $\mathcal{X} = (X_1, X_2, \dots, X_r)$ with width $\leq k$. There exists some i_j such that $X_{i_j} \subseteq V(H_j)$ for $j=1, 2$, and 3 , for otherwise $(\mathcal{X} \cap V(H_j))$ is a proper-path-decomposition of H_j with width $\leq k-1$. Without loss of generality, we assume that $i_1 < i_2 < i_3$. It is trivial that $G \setminus V(H_2)$ is a connected subgraph of G . However, $X_{i_1} \cap V(G \setminus V(H_2)) \neq \emptyset$ and $X_{i_2} \cap V(G \setminus V(H_2)) = \emptyset$, and $X_{i_3} \cap V(G \setminus V(H_2)) \neq \emptyset$, contradicting Lemma 2.7. (Notice that a proper-path-decomposition is also a path-decomposition.) Hence, we have $\text{ppw}(G) = k+1$.

Next we show that G is minimal. It is sufficient to show that the proper-path-width of a minor G' of G obtained by deleting or contracting an edge e is at most k .

Case 1: $e \in \{(v_1, v_2), (v_2, v_3), (v_3, v_1)\}$.

Without loss of generality, we assume that $e = (v_1, v_2)$. If G' is obtained by deleting edge (v_1, v_2) , the sequence $(\mathcal{X}_{(1)} \cup \{v_1\}, \{v_1, v_3\}, \mathcal{X}_{(3)} \cup \{v_3\}, \{v_3, v_2\}, \mathcal{X}_{(2)} \cup \{v_2\})$ is a proper-path-decomposition of G' and $\text{ppw}(G') \leq k$. If G' is obtained by contracting edge (v_1, v_2) , the sequence $(\mathcal{X}_{(1)} \cup \{v_1\}, \mathcal{X}_{(2)} \cup \{v_1\}, \{v_1, v_3\}, \mathcal{X}_{(3)} \cup \{v_3\})$ is a proper-path-decomposition of G' and again $\text{ppw}(G') \leq k$.

Case 2: $e \notin \{(v_1, v_2), (v_2, v_3), (v_3, v_1)\}$.

Without loss of generality, we assume that $e \in E(H_2)$. G' is a delta-composition of H_1, H'_2 , and H_3 , where H'_2 is a minor of H_2 obtained by deleting or contracting e .

Claim. *There exists a proper-path-decomposition $\mathcal{X}^* = (X_1^*, X_2^*, \dots, X_{r'}^*)$ of H'_2 with width $\leq k-1$ such that $v_2 \in X_a^*$ and $|X_a^*| \leq k-1$ for some a .*

It follows from the claim that the sequence $(\mathcal{X}_{(1)} \cup \{v_1\}, X_1^* \cup \{v_1\}, \dots, X_{a-1}^* \cup \{v_1\}, X_a^* \cup \{v_1, v_3\}, X_{a+1}^* \cup \{v_3\}, \dots, X_{r'}^* \cup \{v_3\}, \mathcal{X}_{(3)} \cup \{v_3\})$ is a proper-path-decomposition of G' and $\text{ppw}(G') \leq k$.

It remains to prove the claim. Let $\mathcal{X}' = (X'_1, X'_2, \dots, X'_{r'})$ be a proper-path-decomposition of H'_2 with width $\leq k-1$. We assume, without loss of generality, that $X'_i \not\subseteq X'_j$ for any distinct i and j , and $|X'_i| = k$ for any X'_i that contains v_2 . Assume first that $v_2 \in X'_1$ (resp. $v_2 \in X'_{r'}$). Then $(\{v_2\}, \mathcal{X}')$ (resp. $(\mathcal{X}', \{v_2\})$) is a required sequence. Assume next that $v_2 \in X'_p \cap X'_{p+1}$ for some p ($1 < p < r'-1$). Since $|X'_p| = |X'_{p+1}| = k$ and $X'_p \neq X'_{p+1}$, $|X'_p \cap X'_{p+1}| \leq k-1$. Thus, $(X'_1, \dots, X'_p, X'_p \cap X'_{p+1}, X'_{p+1}, \dots, X'_{r'})$ is a required sequence. Assume finally that $v_2 \in X'_p$ and $v_2 \notin X'_{p-1} \cup X'_{p+1}$ ($1 < p < r'$). If $|X'_{p-1} \cap X'_p| = k-1$ then $(X'_{p-1} - \{v_2\}) \subseteq X'_{p-1}$, since $v_2 \notin X'_{p-1}$ and $|X'_p| = k$. Similarly, if $|X'_{p+1} \cap X'_p| = k-1$ then $(X'_p - \{v_2\}) \subseteq X'_{p+1}$. Hence, if $|X'_{p-1} \cap X'_p| = |X'_{p+1} \cap X'_p| = k-1$ then $|X'_{p-1} \cap X'_p \cap X'_{p+1}| = |X'_p - \{v_2\}| = k-1$, contradicting Definition 3.1. Thus, $|X'_{p-1} \cap X'_p| \leq k-2$ or $|X'_p \cap X'_{p+1}| \leq k-2$. Then $(X'_1, \dots, X'_{p-1}, (X'_{p-1} \cap X'_p) \cup \{v_2\}, X'_p, \dots, X'_{r'})$ is a required sequence for the former case, and $(X'_1, \dots, X'_p, (X'_p \cap X'_{p+1}) \cup \{v_2\}, X'_{p+1}, \dots, X'_{r'})$ for the latter. This completes the proof of the claim. Thus the proper-path-width of any proper minor of G is at most k , and $G \in \Omega(\mathcal{P}_k)$. $\square$

Notice that the above theorem does not hold for $\Omega(\mathcal{F}_k)$. Although a graph shown in Fig. 4 is a delta-composition of graphs in $\Omega(\mathcal{F}_1)$, it is not in $\Omega(\mathcal{F}_2)$ because its minor shown in Fig. 5 is in $\Omega(\mathcal{F}_2)$. Notice also that the star- and delta-compositions are not sufficient to characterize minimal forbidden minors for $\mathcal{P}_k$. A graph in $\Omega(\mathcal{P}_2)$ shown in Fig. 6 is neither a star-composition nor a delta-composition of graphs in $\Omega(\mathcal{P}_1)$.

We conclude with the following remarks: Similar results can be found in the literature. Parsons [11] gave a recursive forbidden subgraph characterization of trees with the edge-search number more than k , for each $k \geq 1$, by proving a similar result as

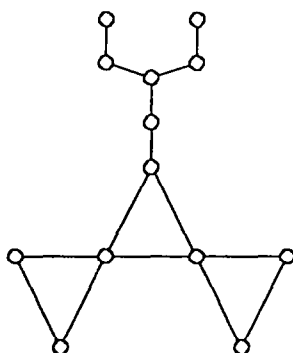


Fig. 4. A graph not in $\Omega(\mathcal{F}_2)$ that is a delta-composition of graphs in $\Omega(\mathcal{F}_1)$.

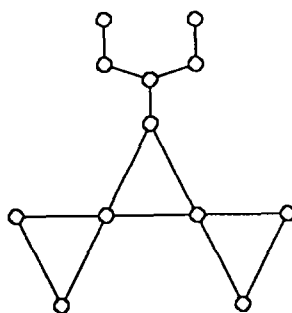


Fig. 5. A graph in $\Omega(\mathcal{F}_2)$.

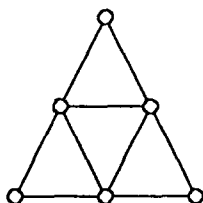


Fig. 6. A graph in $\Omega(\mathcal{P}_2)$.

Lemma 2.10. Möhring [10] mentioned without proof that similar results can be obtained for the node-search number. We learned recently that Scheffler [16] obtained independently the same result as Lemma 2.10 in this paper. Scheffler also gave a linear-time algorithm to determine the path-width of a given tree. We can also give a linear-time algorithm to determine the proper-path-width of a given tree [18]. A special case of Theorem 3.3 when $k=2$ was proved by Takeuchi et al. [20] and independently by Fukuhara [8]. Relations between the path-width, node-search number, and interior k -caterpillars are mentioned in [10, 18], and the proper-path-width, mixed-search number, and k -paths are mentioned in [17].

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